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REVIEW PAPER

Optimisation of shovel-truck haulage system in an open pit using queuing approach

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Abstract
Open-pit mining operations involve loading and hauling of material from the pit to the processing plant or waste dump depending on the ore grade. These two activities account for more than 50% of total operating costs of the mine. There are several loading and hauling systems that are employed depending on the geometry of the ore body. This study applies a shovel-truck haulage system in a limestone open-pit mine to optimise the productivity of the plant using queuing theory technique. This study developed a model using Mat-lab software to calculate the inter-arrival rate and service rate for different numbers of trucks subjected to the same haulage system. The current system used at the limestone cement plant quarry has an inter-arrival rate of 24 trucks/hour and a service rate of 10 trucks/hour with 16 trucks and 2 shovels in the system. Using this data on the developed model, it is evident that when the number of trucks increases to 12, the system daily production increased up to an optimal point of 7644.52 tonnes/day. Thereafter, an increase in the number of trucks reduced the truck productivity, hence reducing the system production and increasing the cost per tonne of material hauled.

Keywords Haulage system · Inter-arrival time · Service rate · Optimisation · Waiting times · Queuing approach

Introduction

Surface mining is the most practiced mining method carried out across the world. In surface mining, there are five stages which include prospecting, exploration, development, exploitation and reclamation (Hartman and Mutmansky 2002). Surface mining can be defined as either open-pit mining, open cast mining, strip mining, dredging or mountaintop removal. Open-pit production operations account for more than 60% of all surface mining production (Hartman and Mutmansky 2002). The production operation in open-pit mining involves a series of activities from the loading, travelling loaded, manoeuvring at the dump, dumping and travelling back to the loader (Fisonga and Mutambo 2017). Loading and haulage are the major activities for the mining transport scheme, accounting for more than 50% of total operating costs (Alarie and Gamache 2002). These operation activities are commonly carried out using shovels and trucks due to their system flexibility in different operation conditions (Nehring et al. 2018).

In shovel-truck haulage system, there are waiting times experienced at loading and dumping points. The waiting times at these stations reduce the efficiency of operation and hence increase the unit cost per material hauled (El-Moslmani 2002). Queuing theory is one of the approaches commonly used in shovel-truck production optimisation as it provides services for the randomly increasing demands (Trivedi et al. 1999). The approach gives a better method of estimating the waiting times in a haulage system. This is because the approach is computationally fast and quite simple to formulate as compared to simulation approaches. This enables forward dispatch of important information in the best and quickest way possible (Jorgen 1979).

There are two types of channel of service in queuing theory: single-server channel and multi-server channel. In a single-server channel, the trucks align to one loader in the system. Figure 1 represents the single-server channel system whereby the trucks are represented by the circle, and the server is

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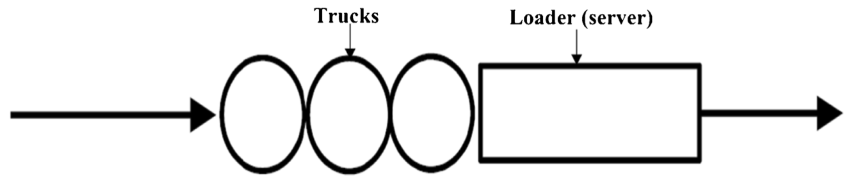

Fig. 1 Single-server channel system *(*Meredith 2013*)*

represented by the rectangle. The trucks have to queue to wait for the service if it is not immediately available until when they are loaded and then proceed through the system once they have been served.

In the multi-server channel, the loaders are arranged in parallel and trucks queue at each loader as shown in Fig. 2a, or a single queue lining for multiple loaders where the next truck in the service line goes to the immediately available loader as shown in Fig. 2b. Multi-server channel is the most used system in the day-to-day life with the single queue for multiple loaders being common since it is more efficient in providing services (Meredith 2013).

Koenigsberg (1958) was the first to apply queuing theory in a mining operation by modelling conventional, mechanised room and pillar mining operations for a closed-loop queuing system. The system considered a finite number of mine faces based on the assumption of exponential service time distributions (Koenigsberg 1958). Although simulation was common around the 1960s, the queuing approach grew in demand since computer simulation by then needed an expensive large computer memory and consumed more CPU (central processing unit) time. Consequently, analytical modelling approaches like queuing theory with little or no computing requirement become a viable alternative to computer simulation models (Meredith 2013).

Karshenas (1989) modified the queuing approach for equipment selection with the study being continued by (Huang and Kumar 1994) in an attempt to develop a model for selecting truck fleet size using a more accurate productivity estimate to minimise the cost of idle equipment. El-Moslmani (2002) developed a queuing model that used server utilisation in the calculation of haulage system production. The computer model was called FLSELECTOR, and it was used to determine the best fleet size. The model allowed the optimum fleet to be selected based on the least cost and maximum production (El-Moslmani 2002).

Krause and Musingwini (2007) showed that a modified "Machine Repair Model," which is an example of a finite source queuing model could also be applied in the estimation of fleet size and yielded accurate results. In the study, the equations of the Machine Repair Model were adjusted to fit loading and hauling situations, while the average time of the trucks waiting for repair becomes the average amount of time a truck can queue at either dumpsite or loading site (Krause and Musingwini 2007). Hai (2016) used single-channel queuing model to determine the relationship between different numbers of trucks with shovel utilisation, system production and the length queue. The model could be used in any haulage system if data on the arrival times of the trucks and service times of the shovel fit to the exponential distribution (Hai 2016).

This paper focuses on the single queue multi-server channel as an approach to optimise the shovel-truck haulage system in open-pit mining. A queuing model is developed in Mat-lab software and tested using the data collected from the limestone quarry. The output from the model is used in equipment selection by comparing the performance of the current system with a combination of different fleet sizes under the same working conditions.

Methodology

Queuing theory is an approach of system performance analysis that helps the mining engineer to decide on the improvement of the mining system. Queues in surface mining haulage system are experienced when trucks line up for loading or dumping. This study employs multichannel queuing approach ($M_1/M_2/S/n$: FCFS) to develop the model, whereby M_1 and M_2 represent the probability distribution of truck inter-arrival time and the probability distribution of service time respectively. The servers are represented by S; fleet size is denoted by n and (FCFS) represents the first-come first-served service discipline. The model is developed under the following assumptions (Hai 2016):

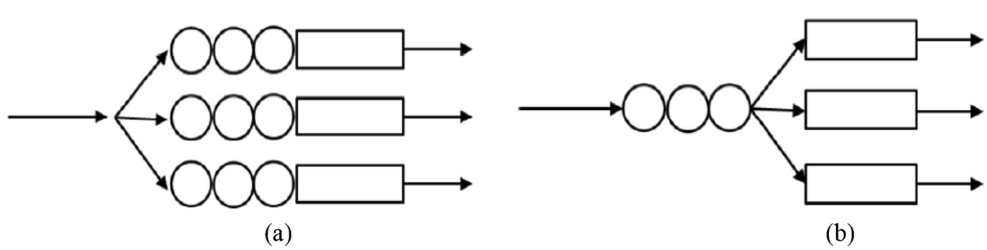

Fig. 2 Multi-server channel system *(*Meredith 2013*)*

- Poisson arrivals for the trucks
- Negative exponential service time for the loaders
- Discipline order of first come, first served (FCFS)
- Equal mean service rate for all loaders (homogenous loaders)
- Homogenous fleet for all the shovels (servers)
- A finite fleet size which in the model corresponds to K

A multichannel queuing model developed captures the activities and predicts the behaviour of the haulage system from loading at the shovel to dumping at the crusher and back at the loading points. The trucks' inter-arrival time (min), service time (min), number of loaders, the truck capacities and the current number of trucks in the system are recorded and analysed based on the assumptions of a multichannel queuing approach. The data collected for the trucks' inter-arrival time (min) and service time (min) must have a negative exponential distribution for it to be run in the queuing model. Mat-lab distribution fitting application is used to check both the inter-arrival time and service time distribution which gives the probability density function of the data against time.

The haulage system consists of loaders and trucks used in the material handling in the mine. The data collected in the mine is the truck inter-arrival time and loader service time in a stable working shift. This data is used to calculate truck inter-arrival rate, λ, and loader service rate, μ. These parameters are used to define system utilisation ρ which is the measure of traffic congestion as expressed in Eq. 1 (Meredith 2013).

$$\rho = \frac{r}{s} = \frac{\lambda}{s\mu}, \tag{1}$$

where r is the ratio between inter-arrival rate and service rate for a single loader and s is the number of shovels.

When $\rho > 1$, it means that the average truck arriving in the scheme exceeds the average shovel service rate ($\lambda > s\mu$). When $\rho < 1$, it means the shovel service rate exceeds average trucks' arrival rates. In conditions where $1 < \rho$, the possibility of having zero trucks to the system is given by probability expression in Eq. 2 (Trivedi et al. 1999) as:

$$Po = \left\{ \sum_{n=0}^{s-1} \frac{K!}{(K-n)!} r^n + \sum_{n-s}^{K} \frac{K!}{(K-s)!s!s^{n-s}} r^n \right\}^{-1}. \tag{2}$$

Alternatively, the possibility of having 'n' trucks in the system is given by probability expression in Eq. 3 *(*Trivedi et al. 1999*)* as:

$$Pn = \begin{cases} \binom{K}{n} r^n P_0 & n = 0, 1, \dots, s-1 \\ \dfrac{K!}{(K-n)!s!s^{n-s}} r^n P_0 & n = s, s+1, \dots, K. \end{cases} \tag{3}$$

where K is the fleet size.

Hence, the expected number of trucks, L_q, in the service line based on the probability, P_n, is given by Eq. 4 (Trivedi et al. 1999).

$$L_q = \sum_{n=s}^{K} (n-s)P_n. \tag{4}$$

The expected number of trucks in the system, L_s, and the average time truck spends in the queue, W_q, are calculated by applying Little's formula. Hence, the expected number of trucks in the system is given in Eq. 5 (Shortle et al. 2018):

$$L_s = \sum_{n=0}^{K} nP_n. \tag{5}$$

The long-term average number of units in a stable system, L_s, is equal to the product of long term average effective arrival rate $(\overline{\lambda})$ and the average time a truck spends in the system, W_s. The effective arrival $\overline{\lambda}$ is given by Eq. 6 (Trivedi et al. 1999):

$$\overline{\lambda} = \sum_{n=0}^{K} \lambda(K-n)P_n. \tag{6}$$

The expected time a truck will spend in the queue is given by Eq. 7 (Trivedi et al. 1999):

$$W_q = \frac{L_q}{\overline{\lambda}}. \tag{7}$$

Likewise, the average time a truck spends in the queuing system, W_s, can be expressed as in Eq. 8 (Trivedi et al. 1999):

$$W_s = \frac{L_s}{\overline{\lambda}}. \tag{8}$$

The utilisation of the shovel, η_s, and the utilisation of the trucks, η_t, are given in Eqs. 9 and 10, respectively (Trivedi et al. 1999).

$$\eta_s = 1-P_0. \tag{9}$$

$$\eta_t = 1-\frac{W_q}{W_q + cycle\ time} \tag{10}$$

The cycle time includes loading time, travelling when empty, dumping time, travelling when empty and manoeuvring time at the loading and dumping points. System production is the parameter of prime importance to a mining company. This is because the more the material delivered to the processing plant, the higher the profit generated at a given time. The hourly system production is given as shown in Eq. 11 (Hai 2016):

Table 1 Shovel service time and trucks' inter-arrival time data

Service time (min)	Inter-arrival time (min)	Service time (min)	Inter-arrival time (min)
7.35	0.78	7.44	4.08
4.01	2.13	5.38	0.78
6.37	3.53	7.41	0.74
5.51	3.65	7.02	2.63
6.56	1.80	7.49	1.15
6.50	1.78	7.12	3.14
6.21	3.88	7.02	0.55
4.49	2.49	6.32	0.76
6.29	8.67	6.13	2.03
5.49	2.47	6.40	1.45
6.46	2.81	9.01	0.88
5.46	4.00	6.02	1.54
5.58	4.28	6.40	0.41
7.00	4.74	5.78	0.59
6.07	6.07	6.21	0.75
6.14	4.18	4.51	2.32
6.26	1.32	7.00	2.25
5.37	3.63	8.01	0.04
6.43	6.70	7.11	2.78
4.42	4.85	6.22	0.28
5.43	1.99	6.41	1.10
5.41	3.20	5.49	0.81
7.20	5.37	6.34	1.92
7.27	3.80	6.29	0.89
6.51	0.97	6.05	1.86

$$\begin{aligned} System\ Production = {} & Service\ rate \times shovel\ utilisation \\ & \times number\ of\ shovel \\ & \times truck\ capacity \end{aligned} \tag{11}$$

Fig. 3 Bucket capacity shovels of 3.5 m^3 loading into 25 tonnes capacity trucks at CataPutt quarry

The total hourly operation cost of the system is given by, $C_1S + C_2N$ (Hai 2016), where C_1 is the cost per unit time of the shovel and C_2 is the cost per unit time of the truck. S represents the number of shovels, and N represents the number of trucks. The total cost of unit production can be expressed as Eq. 12 (Hai 2016):

$$Total\ Unit\ Cost = \frac{C_1S + C_2N}{System\ Production} \tag{12}$$

After deriving the cost model to determine the unit production cost for different numbers of trucks, the total unit cost is plotted against the number of trucks, and the optimum fleet size is determined as the number of trucks corresponding to the minimum cost in the plotted graph (Ercelebi and Bascetin 2009). This is illustrated in "Results" (Fig. 11), where the unit cost is plotted against the number of trucks. The graph shows the optimal point at which the number of trucks is having a minimum cost of operation.

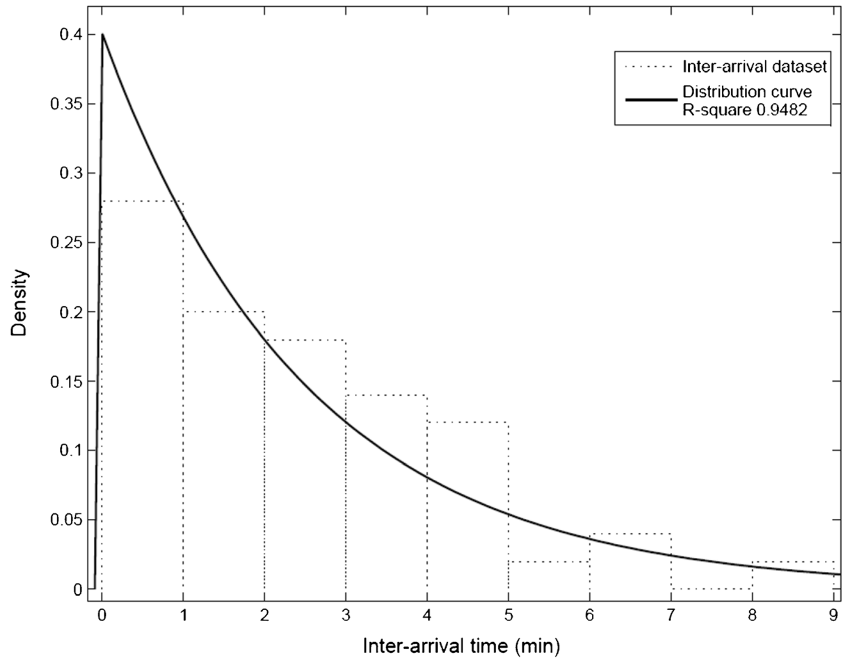

Fig. 4 Inter-arrival time distribution

Results

Case study application in mining

The quarry used in the case study is CataPutt quarry located in Kilifi County, Kenya. The quarry is 6 km from the processing plant. The loaders' service time and trucks' inter-arrival time data were collected from the shovels and trucks operating in the quarry for the system shown in Fig. 3. The data collected in the quarry was tabulated as shown in Table 1.

The data was first cleaned by determining the outliers, that is the time values greater than $Q_3 + 1.5$ (interquartile range) or lower than $Q_1 - 1.5$ (interquartile range). Q_3 and Q_1 are the third quarter and the first quarter, respectively, of the data arranged in ascending order.

The queuing model was coded in Mat-lab software according to the queuing theory principles as shown in Eqs. 1–12. After developing the model, the data collected in a limestone quarry was fitted in the Mat-lab software distribution application. The model was tested using secondary data that was

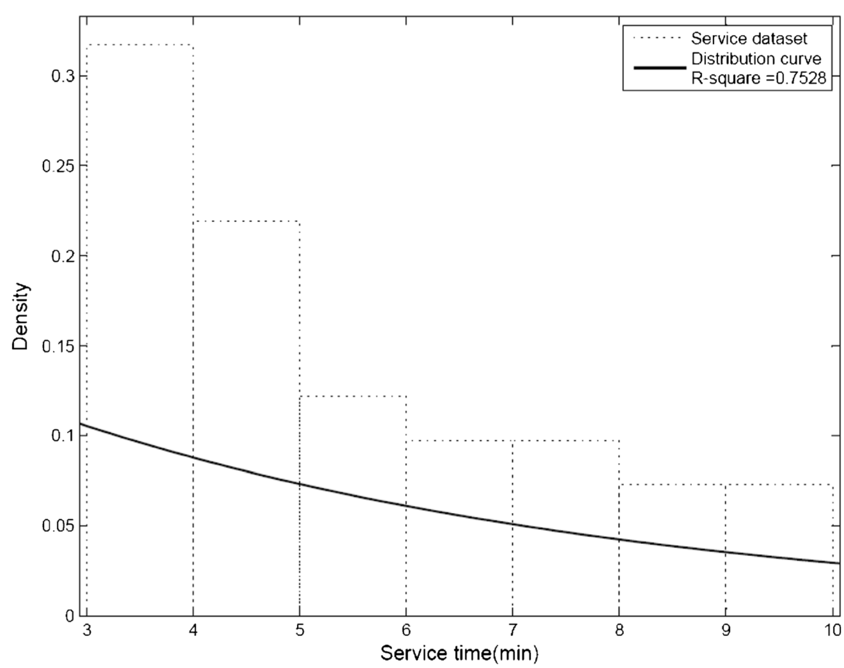

Fig. 5 Service time distribution

Table 2 Model input parameters

Input parameters	Amount
Inter-arrival rate, λ	24 trucks/h
Service rate, μ	10 trucks/h
Number of shovels, S	2
Number of trucks, N	16
Loading cost	KShs 24,200 (US$242) per hour
Haulage cost	KShs 17,500 (US$175) per hour

*1US$ = 100 KShs (Kenya shillings)

published by Hai (2016), and it was found to be 97.2% accurate.

Service time and inter-arrival time distribution

The distribution fitting was done in Mat-lab software using a distribution fitting application. Inter-arrival and service times are sorted to create data density, also called the probability density function (PDF) of the continuous random variable. The variable integral across an interval gives the probability of whether the value of the variables lies within the same interval. The graphs in Figs. 4 and 5 show that both inter-arrival time and service time have a negative exponential distribution. The negative exponential distribution explains the time between events in statistical analysis. This can help one determine when the next event in time will likely happen. This means the occurrence of events in time has a negative exponential distribution and thus correlation of the data can be determined by coefficient of determination of the curve. The coefficient of determination ranges between 0 and 1. The higher the coefficient of determination, the closer the collected data is to one another giving high confidence that the data was collected from the same system.

Model input parameters

There are two main parameters of the queuing model: inter-arrival rate and service rate. The input parameters in Table 2 were stored in a Microsoft Excel spreadsheet linked to the model in Mat-lab. The input parameters for haulage system used in this study were generated from 2 shovels and 16 trucks in operation. The inter-arrival rate is calculated from the mean inter-arrival time data, while the service rate is calculated from mean service time data as shown in Eqs. 13 and 14 (Hai 2016).

$$\text{Inter-arrival rate, } \lambda = \frac{60\ (\textit{minutes in an hour})}{\textit{Average inter-arrival time}(\textit{min})} \tag{13}$$

$$\text{Service rate, } \mu = \frac{60\ (\textit{minutes in an hour})}{\textit{Average service time}\ (\textit{min})} \tag{14}$$

Model performance measures

The model performance measures (output) indicate how the system behaves, and using the system performance indicators generated, the system can be readjusted accordingly. Table 3 summarises the performance measures of the system under different fleet sizes in operation from the model developed.

η_s, shovel utilisation (%); η_t, shovel utilisation (%); Q_n, system daily production (tonnes/day); C_t, total unit operation cost (KShs/Tonne)

Table 3 Model performance measures (output)

K	λ	P_0	L_q	L_s	W_q	W_s	η_s	η_t	Q_n	C_t
2	3.00	0.74677	0.00	0.27	0.00	3.14	25.32329	100.00	1935.85	69.40
3	4.51	0.52802	0.01	0.58	0.06	3.20	47.19757	99.88	3608.03	55.98
4	6.01	0.32611	0.08	1.02	0.27	3.41	67.38856	99.43	5151.54	49.26
5	7.51	0.17100	0.31	1.64	0.75	3.89	82.89994	98.41	6337.31	45.24
6	9.01	0.07228	0.85	2.50	1.61	4.75	92.77190	96.56	7091.98	42.55
7	10.51	0.02321	1.75	3.61	2.94	6.08	97.67898	93.73	7467.10	40.63
8	12.02	0.00546	2.92	4.88	4.67	7.81	99.45419	90.05	7602.80	39.19
9	13.52	0.00094	4.19	6.18	6.61	9.75	99.90573	85.93	7637.32	38.08
10	15.02	0.00012	5.46	7.46	8.57	11.71	99.98769	81.74	7643.59	37.18
11	16.52	0.00001	6.69	8.69	10.50	13.64	99.99874	77.65	7644.43	36.45
12	18.03	0.00000	7.88	9.88	12.37	15.51	99.99990	73.66	7644.52	36.14
13	19.53	0.00000	9.04	11.04	14.19	17.33	99.99999	69.77	7644.53	37.87
14	21.03	0.00000	10.18	12.18	15.98	19.12	100.00000	65.96	7644.53	39.89
15	22.53	0.00000	11.30	13.30	17.74	20.88	100.00000	62.22	7644.53	42.12
16	24.03	0.00000	12.41	14.41	19.48	22.62	100.00000	58.52	7644.53	44.45

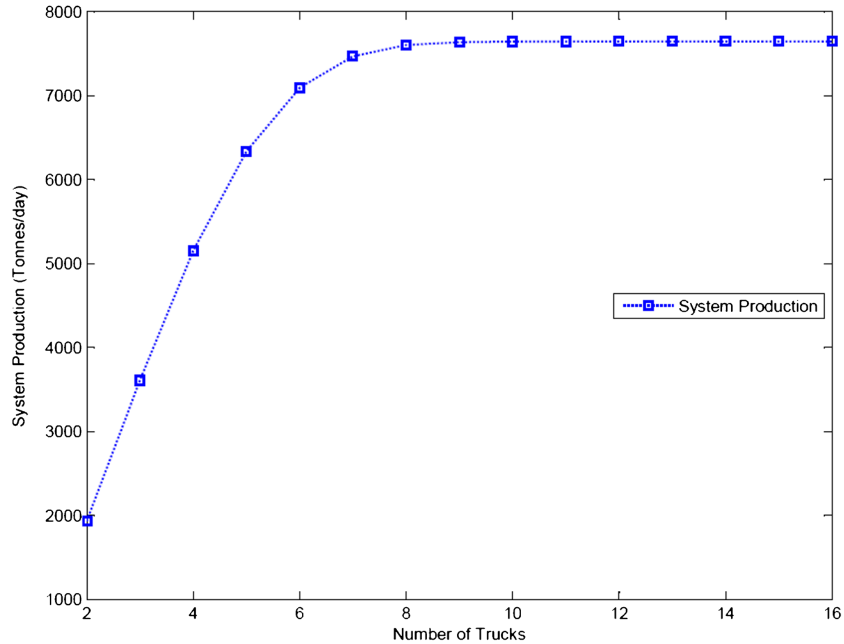

Fig. 6 Graphical representation of system productions against the number of trucks

Discussion

The main output parameters in a mining haulage system are system production, shovel utilisation, trucks utilisation, time trucks spend in the queue and length of queue (Meredith 2013). These performance measures can be integrated to determine the optimal numbers of trucks in the system. The optimality of the system is where the system utilisation and production are increased, while the time trucks spend in the queue, and length of the queue is reduced. This leads to a minimum cost of operation, while achieving the production target (El-Moslmani 2002). Cycle time, availability, utilisation, performance and production efficiencies are important factors when evaluating the fleet size and productivity of a mining operation. The overall efficiency of mining equipment is measured using these input parameters, which will consequently aid in evaluating the operating cost or expenditure of the mine (Samatemba et al. 2020).

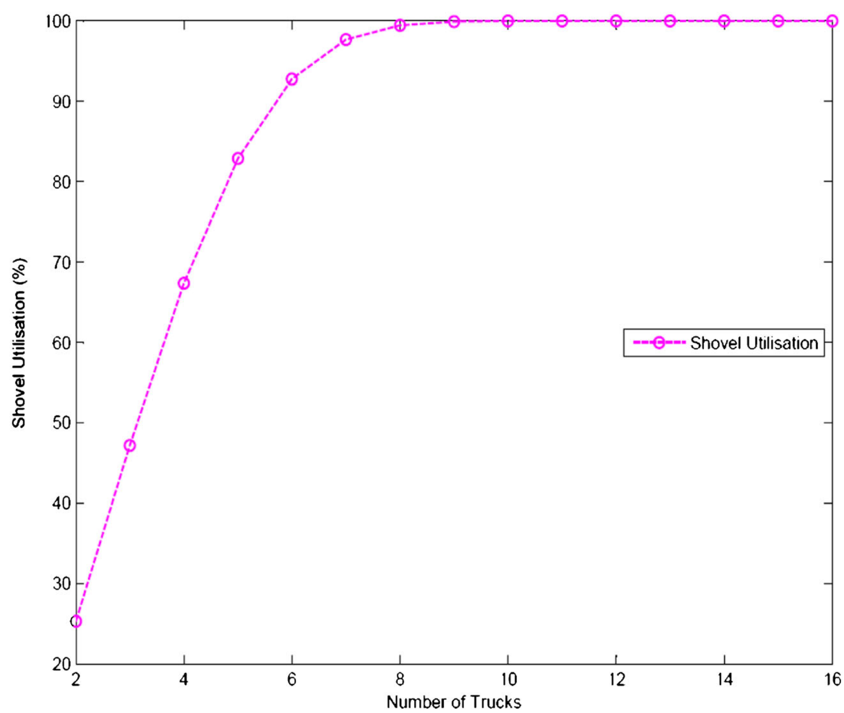

Fig. 7 Graphical representation of shovel utilisation against the number of trucks

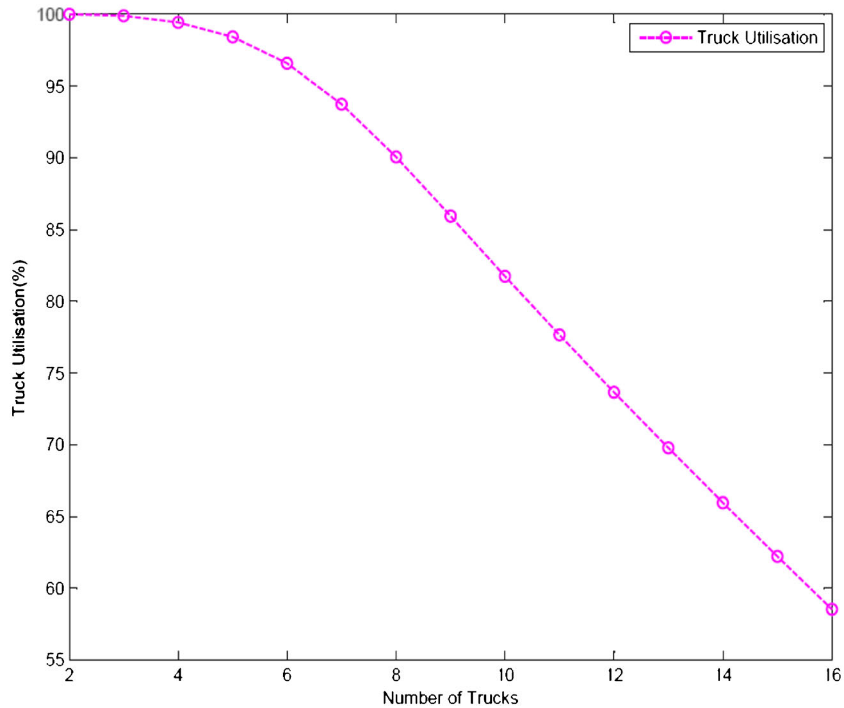

Fig. 8 Graphical representation of truck utilisation against number of trucks

The performance measures are graphically represented in Figs. 6, 7, 8, 9, 10 and 11 indicating the relationship between the number of trucks and the system production, shovel utilisation, trucks utilisation, trucks waiting time in the queue, length of the queue and total unit cost of production. The production of the shovel increases significantly as shown in Fig. 6 as the number of trucks increases to 12 trucks. Afterwards, the curve becomes steady since system optimality has been reached, and the addition of extra trucks leads to the long queue, thus increasing the waiting time due to excessive idling time of the trucks.

The utilisation of the shovel increases significantly as shown in Fig. 7, as the number of trucks increases to 12 trucks. Afterwards, the curve becomes steady since system optimality has been reached, and the addition of extra trucks leads to the

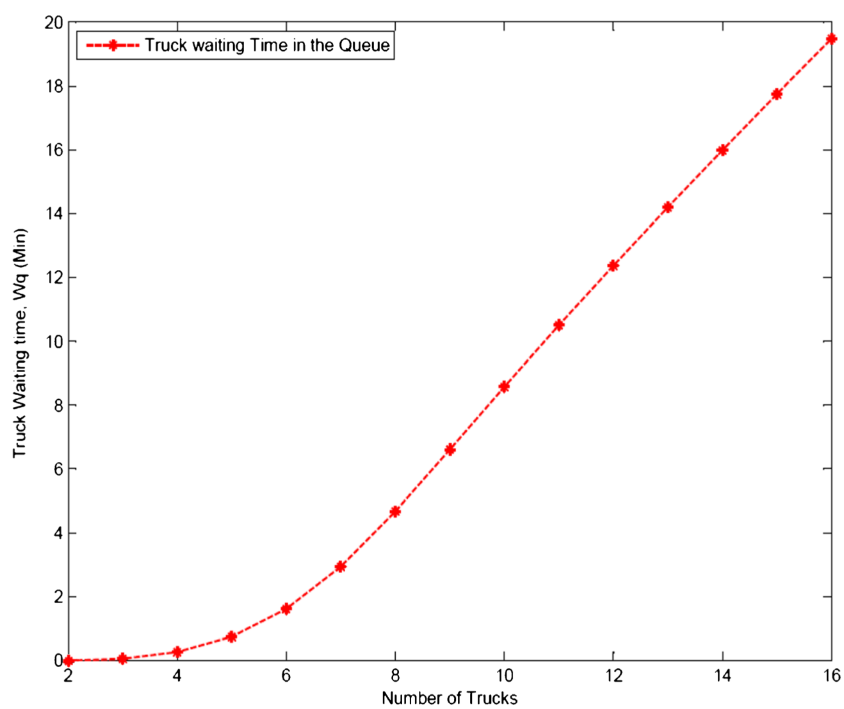

Fig. 9 Graphical representation of trucks waiting time in the queue against the number of trucks

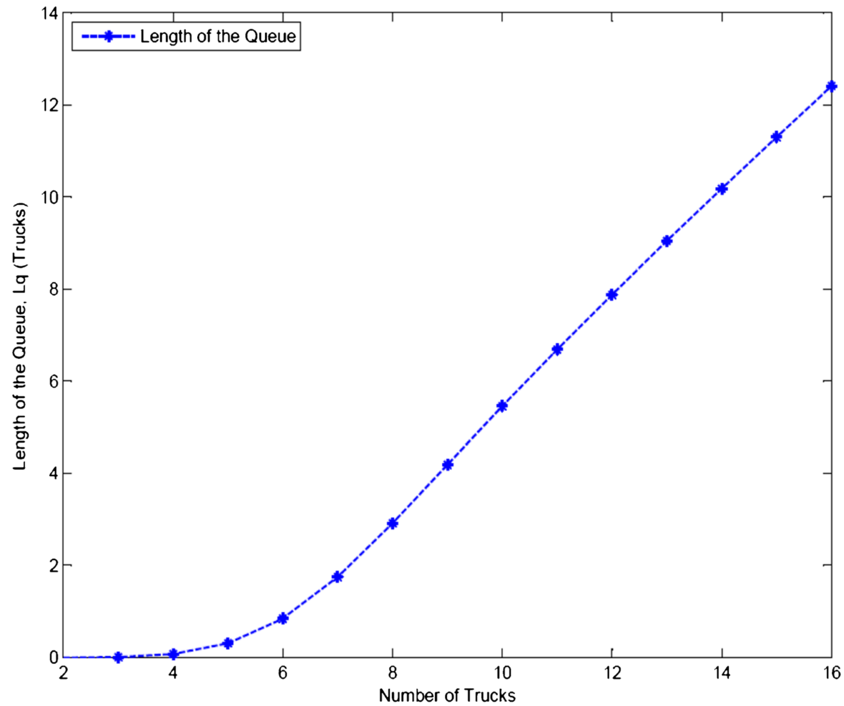

Fig. 10 Graphical representation of the length of the queue against the number of trucks

long queue, thus increasing the waiting time due to excessive idling time of the trucks.

The truck utilisation decreases with an increase in the number of trucks, as seen in Fig. 8, indicating the possibility of a queue being formed, as the number of trucks in the system increase. The increase in the number of trucks in this case means that if a queue is formed, a truck will have to wait longer for the trucks ahead to be loaded.

The truck waiting time in the queue increases as the number of trucks increases as shown in Fig. 9. This means that a truck will have to wait longer for the trucks ahead to get loaded if a queue is formed.

The length of the queue increases as the number of trucks increases as shown in Fig. 10. This means that a truck will have to wait longer for the trucks ahead to get service if a queue is formed.

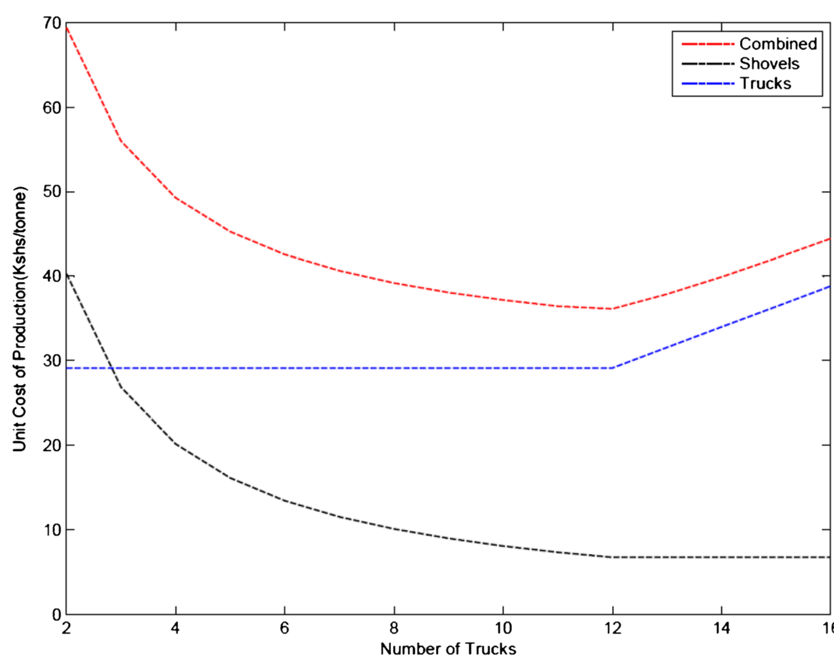

Fig. 11 Graphical representation of the total unit cost of production against the number of trucks

The waiting time in the queue for any system cannot be eliminated, but it can be reduced to save the excessive cost of system operation. The reduction of waiting time is achieved by optimising the system as illustrated in Fig. 11, where the minimum unit cost of production is determined. The unit cost of the trucks remains constant from 2 trucks up to 12 trucks and the shifts upwards. This is because from 2 to 12 trucks, the increase in the number of trucks is proportional to the increase in production, but after 12 trucks, the production remained constant while the number of trucks is increasing.

The unit cost of the shovel decreased from 2 trucks to 12 trucks and then remained constant thereafter. This was because the cost of operation of the shovel remained constant while the production increased with the increasing number of trucks, but after 12 trucks, the cost of operation and system production remained constant giving a horizontal curve. The optimisation curve which is named the 'combined curve' as shown in Fig. 11 is the total unit cost for the trucks and shovels. The total unit cost reduces with the increased number of trucks up to 12, and then it significantly shifts upwards. The minimum total unit cost of production is achieved with 12 trucks. This indicates the null point that represents the number of trucks that can operate optimally with the two designated shovels to meet the company's production target at minimum cost.

Conclusion

The current study focused on the application of the queuing approach to optimise shovel-truck haulage operations at a limestone open-pit mine in Kilifi County, Kenya. The shovel-truck system used in the model had 2 shovels and 16 trucks in operation. The results from the model indicated that the waiting times were long as a result of an excess number of trucks in the system. It is also established that the waiting time at the queue, Wq, and length of the queue, Lq, kept on increasing as the number of trucks increased. In order to have an optimised shovel-truck haulage system, it requires the reduction of the waiting times, consequently, reducing the length of the queues while attaining production targets at minimum costs.

The optimal number of trucks from the model is found to be 12, and thus, the 4 extra trucks should be kept on standby to be used when the trucks in operation have either broken down or are undergoing maintenance. Shovel utilisation was at its peak with 12 trucks. This reduced the waiting time a truck takes in the queue by 7 min 6.6 s as seen in Table 3. This figure is the difference in waiting time for 16 and 12 trucks.

The multichannel queuing model ($M_1/M_2/S/n$) proved to be a good approach to optimise the shovel-truck haulage system in mining. This approach gives a variety of performance measures that can be used to readjust the system to operate optimally. The model uses the input parameters to measure performance in relation to the number of operating trucks and shovels. This interrelationship gives a clear vision of the adjustments that can be made in the system to achieve the minimal operation costs.

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Declarations

Conflict of interest The authors declare that they have no competing interests.

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